

Second Terminal Examination - 2079

Class : XII

Subject: Mathematics (0081)

Full Marks: 75

Time: 3 hrs.

SET 'B'

Pass Marks: 30

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all the questions.

Group 'A'

Rewrite the best alternative of each question in your answer sheet. [11 × 1 = 11]

1. How many ways can letters of the 'COMMITTEE' be arranged?

a) $\frac{9!}{(2!)^2}$

b) $\frac{9!}{(2!)^3}$

c) $\frac{9!}{2!}$

d) $9!$

2. ${}^8C_0 + {}^8C_1 + {}^8C_2 + \dots + {}^8C_8 =$

a) 256

b) 512

c) 128

d) 64

3. $\frac{a+b\omega+c\omega^2}{a\omega+b\omega^2+c} =$

a) 1

b) ω

c) ω^2

d) 0

4. $\sin (\cot^{-1} x) =$

a) $\sqrt{1+x^2}$

b) x

c) $(1+x^2)^{-\frac{3}{2}}$

d) $(1+x^2)^{-\frac{1}{2}}$

5. A conic having eccentricity greater than one is

a) ellipse

b) hyperbola

c) parabola

d) circle

6. The distance of the point (3, 4, 2) from the x -axis is

a) 3

b) 4

c) $\sqrt{29}$

d) $\sqrt{20}$

7. If $\vec{a} = 2\vec{i} - 3\vec{k}$ then $\vec{a} \times \vec{a}$ is
 a) $4\vec{i} - 9\vec{k}$ b) zero c) a zero vector d) 13
8. Two cards are drawn one after another without replacement. What is the probability that the second card is a king if the first card is not a king?
 a) $\frac{1}{13}$ b) $\frac{1}{17}$ c) $\frac{4}{13}$ d) $\frac{4}{51}$
9. If $f(x) = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$ to ∞ , then $f'(x) =$
 a) e^x b) $\ln x$ c) e^{-x} d) $e^x - 1$
10. $\int \frac{dx}{\sqrt{9+x^2}} =$
 a) $\sin^{-1}x + c$ b) $\sin^{-1}\left(\frac{x}{3}\right) + c$
 c) $\cos^{-1}\left(\frac{x}{3}\right) + c$ d) $\ln(x + \sqrt{9+x^2}) + c$
11. During the process of solving system of linear equations in x, y, z , we obtain $0 \times z = 1$ at the end of forward elimination. The system is
 a) inconsistent and has no solution.
 b) consistent and has unique solution.
 c) inconsistent and has infinitely many solutions.
 d) consistent and has infinitely many solutions.

Group 'B'

Short Answer Questions.

[8 × 5 = 40]

12. a) How many four digits odd numbers can be formed by using the digits 0, 1, 2, 3, 4, 5 only once? [2]

- b) In how many ways can a football team of 11 players be formed from 16 players? How many of these will be formed if 2 particular players are always included but one particular player is always excluded ? [1+2]
13. a) Find the coefficient of x^6 in the expansion of $\left(x^3 + \frac{1}{x}\right)^{10}$. [2]
 b) Prove that: $1 + \frac{2^2}{2!} + \frac{3^2}{3!} + \frac{4^2}{4!} + \dots$ to $\infty = 2e$ [3]
14. a) Find the equation of the ellipse in standard position with focus at $(0, -2)$ and a vertex at $(0, 5)$. [2]
 b) Find the eccentricity and foci of the hyperbola $5x^2 - 20y^2 - 20x = 0$. [3]
15. a) Prove that: $\cos^{-1} x = 2 \cos^{-1} \sqrt{\frac{1+x}{2}}$ [2]
 b) If $\tan^{-1}x + \tan^{-1}y + \tan^{-1}z = \frac{\pi}{2}$, show that $xy + yz + zx = 1$. [3]
16. a) If α , β and γ are the direction angles of a line, prove that $\sin^2 \alpha + \sin^2 \beta + \sin^2 \gamma = 2$ [2]
 b) Solve the system $x - 3y = 5$, $-x + y = 9$ using row-equivalent matrix method. [3]
17. a) Find the area of a triangle whose two sides are $\vec{i} + 2\vec{j} + 3\vec{k}$ and $-3\vec{i} - 2\vec{j} + \vec{k}$. [2]
 b) Prove vectorically that : $\sin(A-B) = \sin A \cos B - \cos A \sin B$. [3]
18. a) Integrate: $\int \frac{dx}{e^x + e^{-x}}$ [2]
 b) Prove that $\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln(x + \sqrt{x^2 - a^2}) + c$ [3]
19. Using simplex method, maximize $P = 8x + 10y$ subject to $x + 2y \leq 30$, $x + y \leq 20$, $x, y \geq 0$. [5]

Group 'C'

Long Answer Questions.

[5 × 8 = 24]

20. a) Out of 32 students in a class, 8 students are girls. If 3 students are selected, find the probability that 2 students are boys and one is girl. [2]
- b) A company produces electronics chips by a process that manufactures normally average 20% defective products. A sample of four chips is selected at random and tested, what is the probability that
- (i) at most one chip is defective.
- (ii) more than one chip are defective. [3]
- c) Solve the following equations using Gauss-Elimination method:
 $x - 2y + 3z = 2$, $2x - 3y + z = 1$, $3x - y + 2z = 9$. [3]
21. a) State De-Moivre's theorem. Apply it to solve $z^3 = -1$. [5]
- b) Find the equation of the plane through the point $(-2, 3, 4)$ and perpendicular to the planes $3x + 2y + 2z - 8 = 0$ and $2x + 3y + 4z - 6 = 0$. [3]
22. a) Find the equation of the tangent to the curve $y = x^2 + 3$ at $x = 1$. [2]
- b) Using L Hospital's rule, evaluate: $\lim_{x \rightarrow 0^+} \frac{\log(\tan x)}{\log x}$. [3]
- c) State Mean Value Theorem. Given that a function $f(x) = x^2 - 4x + 3$ satisfies the theorem in $[1, 4]$, find a point on the curve where tangent is parallel to the chord joining the points at $x = 1$ and $x = 4$. [1+2]

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